

Detecting Natural Emotions in Virtual Reality Through Facial Movement Analysis

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ABSTRACT

Facial expression recognition (FER) plays a key role in enabling emotionally responsive systems in virtual reality (VR). However, most existing FER datasets in VR are constructed from acted emotions, which may not accurately reflect how emotions are naturally expressed. Prior work suggests that natural emotions differ from acted ones in terms of intensity, subtlety, and temporal dynamics. This mismatch raises concerns about the real-world performance of FER models trained solely on acted data.

In this paper, we investigate facial emotion recognition in VR using natural expressions captured from participants watching emotionally evocative videos, alongside acted expressions performed by the same individuals. We compare the two types of data to analyze differences in Action Unit (AU) activations and their impact on automatic emotion recognition.

Our analysis across the two types of emotion data collections shows that natural expressions activate different AUs than acted emotions and exhibit greater temporal variability. We then evaluate a machine learning model trained on this data and show that natural expressions are indeed more difficult to recognize, yielding an average F1-Score of 73%, compared to 81% for acted emotions.

1 INTRODUCTION

Emotion recognition aims to infer a person's emotional state from observable cues, such as facial expressions, vocal tone, or physiological signals (e.g., electroencephalogram). In VR environments, emotion recognition enables more immersive and responsive interactions by allowing systems to adapt to users' affective states in real time [37]. This capability supports a range of applications, including emotionally adaptive training simulations [41], social VR experiences [27], mental health monitoring [33], and personalized gameplay [6, 28].

Facial expressions arise from underlying muscle movements and offer important cues to emotional states. These can be systematically described using the Facial Action Coding System (FACS), which represents movements as *Action Units (AUs)* [11, 14]. Prior studies have used external cameras to capture facial images and then infer emotional expressions from them [2, 36]. However, such methods are sensitive to lighting, resolution, occlusion, and camera angle, which limits their reliability for detecting subtle, spontaneous emotional responses in naturalistic settings [3].

Facial expression recognition (FER) builds on these cues and is widely used for emotion recognition, including in VR. However, VR headsets cover the upper half of the face, which limits the effectiveness of FER based on external cameras. To address this, some newer headsets include inward-facing infrared cameras that capture facial movements from inside the device, e.g., Meta Quest Pro, and Apple Vision Pro. These systems use the FACS to represent facial muscle activity as AUs, such as inner brow raise. The Meta Quest Pro, for instance, tracks 70 AUs across the upper and lower face, including around the lips (e.g., *LipCornerDepressorLeft*, *LipCornerDepressorRight*) and the nose (e.g., *NoseWrinkleLeft*, *NoseWrinkleRight*).

ACM Reference Format:

Terens Tare, Rahat Rizvi Rahman, Hee Yun Choi, Joonghyo Lim, Go Eun Lee, Seungmoo Lee, Chungyeon Cho, and Kostadin Damevski. 2025. Detecting Natural Emotions in Virtual Reality Through Facial Movement Analysis. In *31st ACM Symposium on Virtual Reality Software and Technology (VRST '25)*, November 12–14, 2025, Montreal, QC, Canada. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3756884.3766013>

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ACM ISBN 979-8-4007-2118-2/2025/11
<https://doi.org/10.1145/3756884.3766013>



Figure 1: Experimental setup showing a participant during passive video viewing for natural emotion elicitation. The right-bottom box shows the content in the VR headset.

Recent studies have started to explore how these AUs can support real-time emotion recognition in VR [25, 26].

Most existing research on VR-based emotion recognition focuses on acted emotions, which are easier to collect under controlled conditions [16, 43]. For example, Jing et al. [16] used emotional event descriptions from the EmoBank corpus that were performed by an actor to generate controlled emotional expressions in a VR setting. In these datasets, participants are instructed to portray specific emotions, such as fear or happiness, while their facial data is recorded. However, prior work has shown that naturally occurring emotions differ substantially from acted ones. Natural emotions are more subtle, less exaggerated and occur more rapidly [21, 24]. Detecting natural emotional responses in VR is essential for enabling more interactive, adaptive, and emotionally responsive virtual environments. However, naturalistic emotional behavior in immersive VR remains significantly underexplored. Previous work has emphasized the importance of studying spontaneous emotion to ensure that emotion recognition systems perform effectively in real-world scenarios [3, 38]. To address this gap, our study focuses on analyzing naturally occurring emotional responses in a VR setting and comparing them with acted expressions recorded using VR headsets.

In this paper, we collect natural facial movement data from individuals wearing a VR headset while they watched emotionally evocative videos. Participants self-report the intensity of the emotions they experience during the videos. To gather a baseline, we subsequently ask each participant to act out the same emotions. Our goal is to understand the differences between natural and acted emotions based on FACS data collected by the VR device. More specifically, we aim to answer the following research questions: (RQ1) How do facial expressions in VR differ between natural and acted emotions? and (RQ2) Are emotions more difficult to recognize based on natural facial expressions than acted facial expressions?

The contributions of the paper are:

- dataset of facial expressions recorded in VR, including both natural and acted emotions, annotated with self-reported emotion intensities and represented using FACS.

- comparative analysis of natural and acted emotional expressions in VR based on FACS and temporal characteristics.
- machine learning approach for emotion recognition in VR using FACS data, evaluated on both natural and acted expressions.

Our results indicate that natural and acted emotional expressions while watching video content in VR differ significantly in their AU activation patterns and temporal dynamics, and that standard emotion recognition models perform substantially worse on natural expressions than on acted ones. We show that models trained on one type of expression generalize poorly to the other, confirming a clear domain gap between natural and acted emotional data.

2 RELATED WORK

We begin by discussing related work on facial expression recognition more broadly. We then focus on facial recognition in VR, including the current state-of-the-art approaches that use the Facial Action Coding System (FACS). Finally, we review prior work that specifically contrasts natural and acted emotions in the context of emotion recognition.

2.1 Facial Emotion Recognition

Facial expression recognition (FER) has become an active area of research, with methods evolving from traditional feature-based approaches to deep learning techniques. Earlier FER methods relied on techniques such as Local Binary Patterns (LBP) [31], LBP on Three Orthogonal Planes (LBP-TOP) [44], Non-negative Matrix Factorization (NMF) [45], and sparse representation learning [46]. These approaches extracted handcrafted features and typically used shallow learning models or basic classifiers for emotion classification.

Adyapady et al. [1] reviewed a wide range of FER techniques and highlighted that while traditional techniques like LBP [19], Gabor filters [4], and optical flow methods showed promise under controlled conditions, their performance often degraded in real-world scenarios. These traditional features were commonly combined with classifiers such as Support Vector Machines (SVMs) and

ensemble-based methods, but their generalization ability remained limited.

In contrast, deep learning-based FER approaches have shown superior performance by automatically learning both spatial and temporal patterns directly from data. Convolutional Neural Networks (CNNs) [17] are widely used to capture spatial features from facial images, while Recurrent Neural Networks (RNNs) [42] help model the changes in facial expressions over time. These models benefit significantly from large-scale annotated datasets. The expansion of benchmark FER datasets such as FER2013 [13] and the Emotion Recognition in the Wild (EmotiW) [7–9] datasets has provided the necessary data to train robust deep learning models. Leveraging these datasets, a variety of deep FER architectures have been proposed, including models by Sun et al. [34, 35], Xie and Hu [40], and Liu et al. [20], achieving state-of-the-art recognition performance.

2.2 Facial Emotion Recognition in VR

VR headsets present unique challenges for emotion recognition due to facial occlusion. The upper half of the face is typically covered, obscuring critical regions like the eyes and brows. This occlusion reduces the effectiveness of conventional vision-based methods, which rely on full facial visibility. For example, Ortmann et al. report 69.8% accuracy on six basic emotions with occluded faces, compared to over 90% on unobstructed data [25].

To address this, recent work has used headset-integrated facial tracking to access internal signals. A recent paper by Ortmann et al. achieved 73% accuracy using facial expression activations (FEAs) captured from inward-facing sensors [26], demonstrating that internal AU measurements can partially recover occluded facial dynamics.

Facial expressions result from specific muscle movements that vary by emotion. The Facial Action Coding System (FACS), introduced by Ekman and Friesen [11], provides a standardized framework for describing these movements as AUs. FACS is widely used for analyzing emotional expressions in a structured and interpretable way [5, 22].

Modern VR headsets can now estimate AUs in real time using inward-facing sensors. These AU values, aligned with the FACS standard, enable emotion recognition without external cameras and support analysis even when parts of the face are occluded. Notable devices include the Meta Quest Pro and Apple Vision Pro, as well as accessories that can be added to VR devices, like the HTC VIVE Facial Tracker and Emteq Pro [39].

The Emteq Pro and HTC VIVE Facial Tracker are accessories that attach to existing VR headsets to capture facial signals. The Emteq Pro provides high-quality electromyography data that can be mapped to AU values, while the HTC VIVE Facial Tracker outputs blendshape data that can be converted into AUs. Both accessories are limited in coverage, since they capture only the lower or the upper part of the face. In contrast, headsets like the Apple Vision Pro integrate inward-facing cameras directly into the device for real-time full-face tracking, though access to raw AU data is still limited.

In this paper, we use the Meta Quest Pro, a standalone VR headset that provides 70 AU measures. This allows us to analyze FACS-aligned AUs of the whole face in real time and compare the dynamics of natural and acted emotions in VR.

2.3 Acted vs. Natural FER

Distinguishing natural from acted emotions has attracted interest from researchers across psychology, computing, and behavioral science. Below, we review prior work that focuses specifically on identifying differences between natural and acted emotion.

A seminal paper that distinguished acted vs. natural emotions in FER is by Mavadati et al. [23], which also curates a novel dataset of natural emotions called DISFA+. There are several key differences between DISFA+ and the dataset in our paper, namely 1) our dataset focuses on VR emotions gathered by inward facing cameras on a VR headset; 2) our dataset uses self-reported emotions, while in DISFA+ the emotions are annotated by a third party; 3) DISFA+ focuses on 12 AUs, while we leverage 70 AUs that are gathered by the Meta Quest Pro VR device.

A survey paper by Jia et al. [15] focuses on methods for detecting whether a facial expression is genuine or posed, using traditional camera-based datasets and feature-driven models. Unlike this paper, it does not perform emotion classification, but instead focuses only on assessing whether the facial emotion is natural or acted.

Ekman et al. focused on the distinction between felt (natural) and false (acted) smiles [12]. They proposed a method for identifying natural smiles based on the presence of specific AUs, particularly *LipCornerPuller*, *CheekRaiser*, and *LidTightener*. Their key finding was that the absence of AU 6 and AU 7 (*CheekRaiser* and *LidTightener*) typically signals a false smile, unless the *LipCornerPuller* is strongly activated, which can obscure the difference. They also noted that false smiles may include slight facial asymmetries, which can aid in detection. Dibeklioglu et al. [10] also focused on distinguishing natural from acted smiles, using an SVM-based model. Like Ekman, they used AU features to train their classifier. Their study found that movements in the lip corners, cheeks, and eyelids are effective indicators for separating natural and posed smiles.

3 DATA COLLECTION

We use AU data provided by the Meta Quest Pro’s inward-facing sensors, which estimate facial muscle movements following the FACS standard, to collect emotion expressions in VR. This section details data collection and experimental design procedures.

We designed the experiment to enable a controlled comparison between natural and acted emotional expressions in a VR environment. Participants wore a VR headset equipped with facial tracking sensors and watched emotionally evocative video clips. While the videos played, the system recorded their facial muscle activity as AU data in real time. After each video, participants reported the type of emotions they felt, along with their intensity and frequency. These self-reports aligned with the facial expression data to support a multimodal analysis of emotional responses.

We structured the study as a within-subjects design, allowing each participant to complete both the natural and acted emotion conditions. In the natural condition, participants passively experienced emotional content. In the acted condition, they intentionally

reproduced the target emotions. This design enabled direct comparison of spontaneous and deliberate expressions within individuals under consistent conditions. Figure 1 illustrates a participant watching emotional content while the system records facial data.

Before beginning the study, each participant signed an informed consent form. We explained that some videos could provoke negative emotional reactions and that facial data would be recorded and used for academic research. The study received IRB approval from Virginia Commonwealth University (Protocol #HM20031878) and took place between March 15 and April 22, 2025.

3.1 Environment Setup

We conducted the study in a soundproof laboratory with stable lighting and minimal visual distractions. Participants sat facing forward, wearing a Meta Quest Pro VR headset. This device includes inward-facing infrared cameras that track 70 facial AUs in real time, capturing values on a continuous scale from 0 to 1. AU data was recorded at 10 frames per second and saved in CSV format, synchronized with video timestamps.

We developed the experimental software in Unity 6 using version 72 of the Meta Quest All-in-One SDK for Unity. Participants viewed video stimuli on the headset display and listened through built-in speakers. During playback, they remained stationary and did not provide any active input.

3.2 Data Collection Procedure

Each session involved one participant and one researcher and followed a fixed experimental sequence. Each session included two main phases: (1) natural emotional response elicitation and (2) acted emotion reenactment.

In the natural phase, participants passively viewed emotional video clips designed to evoke Disgust, Happiness, Anger, Sadness, Fear, Neutral, or Surprise. After each clip, they completed a structured questionnaire via Google Form to report the type, intensity (on a 10-point Likert scale), and frequency of emotions experienced.

Participants could pause or take breaks as needed. The videos used for this part are of different length. While it would have been ideal to work with a standardized set of videos of identical duration, no such corpus exists in this research area. Previous studies have likewise drawn on heterogeneous, naturally occurring materials when standardized stimuli were unavailable. To ensure ecological validity and relevance, we sourced videos from a widely used platform (YouTube), applying consistent selection criteria such as a maximum duration of two minutes and clear alignment with the target emotion. Specifically, the lengths of the videos were as follows: the Anger video was 38 seconds, the Disgust video was 48 seconds, the Fear video was 33 seconds, the Happiness video was 1:49 (min:sec), the Neutral video was 26 seconds, the Sadness video was 1:08 (min:sec), and the Surprise video was 15 seconds. The variance in video length was compensated for by applying filtering phases that retained only the segments where the target emotions were expected, using the methods described in Sections 4.1 and 5.1.

In the acted phase, the name of each target emotion appeared on the headset display, one at a time. Participants then performed facial expressions freely for approximately one minute per emotion.

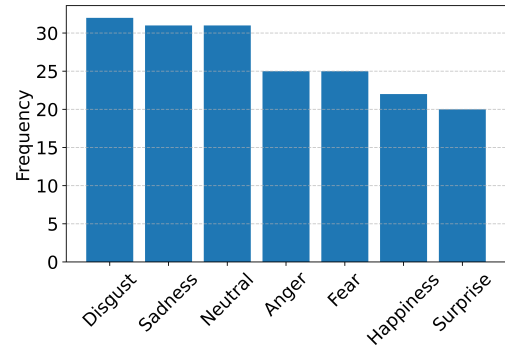


Figure 2: Number of participants reporting each emotion with intensity scores of 5 or above.

The system recorded AU data continuously using the procedure described above. These recordings supported subsequent comparative analysis between natural and acted expressions.

3.3 Collected Data

A total of 34 participants completed the study. Their ages ranged from their 20s to 60s, with an average in the early to mid-30s. Most were university students who volunteered to participate.

The dataset includes multimodal data collected during both the natural and acted conditions:

- (1) Self-Reported Emotional Evaluation Data: Participants reported the type, intensity, and frequency of emotions experienced after each stimulus. Participants could pick from the 7 emotions considered in our study and also specify any other emotion they were experiencing while watching each video. This data was collected via Google Forms and organized into CSV format.
- (2) Behavior-Based AU Time-Series Data: Facial AU data captured by the Meta Quest Pro headset—recorded at 10 frames per second during each emotional induction video and acted emotion session.

Figure 2 shows the frequency distribution of self-reported emotions that match the video’s emotion with intensity scores of 5 or higher. We applied this threshold because the median intensity score across all responses where the emotions matched the video’s emotion was 5. After filtering Disgust was the most frequently reported emotion, experienced by 32 participants. Sadness and Neutral followed with 31 each, while Anger and Fear each appeared in 25 reports. Finally, Happiness was reported by 22 participants and Surprise was reported by 20 participants.

4 RQ1: HOW DO FACIAL EXPRESSIONS IN VR DIFFER BETWEEN NATURAL AND ACTED EMOTIONS?

After completing the data collection phase, we processed and analyzed the AU data to explore the relationship between facial muscle movements and expressed natural and acted emotions. Inspired by a previous study by Namba et al. [24], we trained a logistic regression model to identify which AUs are most predictive of each emotion.

In this analysis, AUs assigned higher model coefficients were interpreted as having a stronger contribution to the expression of the corresponding emotion.

4.1 Data Preprocessing

We began by filtering the data based on the self-reported responses collected during the first part of the experiment. For each video, we excluded any AU recordings in which participants did not report experiencing the target emotion. These instances were not useful for our analysis, as they lacked alignment between video stimulus and perceived emotion.

Next, we reduced the dimensionality of the AU feature space. The Meta Quest Pro tracks 70 facial Action Units, including several pairs of symmetric features (e.g., *EyesLookUpL* and *EyesLookUpR*) that mirror each other. To align with the original FACS specification by Ekman [11], we averaged each symmetric pair into a single feature. This step reduced the total number of features from 70 to 43.

Before training the logistic regression (LR) model, we further processed the AU time-series data to extract more compact and informative representations. We segmented each participant's recordings into 1-second windows and computed the maximum value for each AU within each window. This approach produced one value per second for each AU, emphasizing the most salient moments in the data. Since the AU values already fell within the $[0, 1]$ range, we did not apply additional normalization.

4.2 Logistic Regression Model

To estimate the contribution of each AU to the expression of specific emotions, we trained logistic regression classifiers. Because we aimed to assess the importance of each AU for each emotion independently, we adopted a one-vs-rest binary classification strategy. We trained separate classifiers for each emotion, using both the natural and acted expression datasets. Each model predicted whether a given AU reading corresponded to the target emotion or not. Each model received a 43-dimensional input vector consisting of AU readings and produced a binary output indicating the presence or absence of the target emotion. To fit this binary formulation, we remapped the original seven emotion labels into two classes: positive for the target emotion and negative for all others.

We applied a stratified 80:20 train-test split to preserve the original class distribution. We further applied a stratified 5-fold split on the training set for cross-validation. The best-performing model was then used to identify the Action Units reported in 1. Each model was trained for 50 epochs. To choose an appropriate solver, we compared Adam, Stochastic Gradient Descent (SGD), and Limited-Memory BFGS (LBFGS)[30]. Across all models, LBFGS consistently achieved higher F1-scores than the other solvers. This result is consistent with prior knowledge, as LBFGS is particularly effective for smaller datasets.

4.3 Action Unit Selection

To help distinguish between different emotions and whether an emotion is acted out or not, we aimed to find a reduced subset of AUs that provides a performance close to the full feature set. For this purpose, we used scikit-learn's Recursive Feature Elimination

(RFE) [29]. We ran RFE repeatedly to reduce each logistic regression model into a group of smaller models with several features ranging from 3 to 10. For each subset, we computed the F1-score and selected the feature set that achieved the highest validation score. In the RFE function, we used the logistic regression model's coefficients (feature importance) as an estimator to remove the least important AU. While there can be more complex heuristics that might give better results, for this study, we chose to rely on the base logistic regression coefficients for simplicity. The selected features retained approximately 90% of the original model's F1-score for most cases. The only notable exception was the natural Surprise emotion, which retained 84% of its original F1-score. These results suggest that the selected subsets of AUs provide a compact yet effective representation for both natural and acted emotion classification.

4.4 Results

To answer this research question, we show the most important acted and natural emotion AUs exhibited by the participants in our study in Table 1. The AUs in the table are ordered according to the importance of the logistic regression models.

For comparison, we also show the AUs prescribed by Ekman [11], in no specific order, in the first column of the table. The AUs that are common to Ekman and one of the other columns have a red background, those that are common to natural and acted emotions in our models have a blue background and those that are common to all three columns are denoted in purple background. This color-coding allows us to identify overlaps and discrepancies across the three perspectives. Below, we discuss consistent patterns and notable differences across emotions.

Ekman argued that *UpperLidRaiser* appears with a slight intensity when people feel Surprise. Our models support this association for both natural and acted Surprise. Additional AUs that Ekman linked to Surprise, specifically *InnerBrowRaiser* and *OuterBrowRaiser*, did not appear in the natural or acted expressions in our study. This absence suggests that brow movement does not always accompany Surprise. Instead, we notice the presence of lip-related features such as *LipTightener* and *LipSuckB* on both our natural and acted results. Natural expressions included subtle tongue and lip movements that Ekman did not highlight. Acted expressions, by contrast, showed broader facial engagement such as *CheekPuff*, which likely reflects overacting by the participants, and more involved lip movements, with AUs such as *LipSuckT* and *LipToward* showing up.

For Happiness, Ekman identified *CheekRaiser* and *LipCornerPuller* as commonly observed AUs. Our models also selected *CheekRaiser* as a key indicator for both natural and acted Happiness, reinforcing its role in detecting happy expressions. However, *LipCornerPuller* is notably absent from both models and *JawThrust* is noticed as a common feature instead. In the case of Happiness, the natural model found *LipsToward*, whereas acted Happiness included *LowerLipDepressor*, *LipFunnelerB*, and *CheekPuff*, highlighting a distinction in the facial configurations. Similar to Ekman, natural expressions placed more emphasis on external lip configurations, such as *LipSuckB*, *LipSuckT* and *LipsToward*. In contrast, acted expressions focused more on internal articulator adjustments, particularly involving the tongue.

Table 1: Facial action units (AUs) for natural and acted emotions in our study, compared against Ekman’s defined AUs.

Emotion (Ekman’s AUs)	Natural AUs	Acted AUs
Surprise (<i>InnerBrowRaiser</i> + <i>OuterBrowRaiser</i> + <i>UpperLidRaiser</i> (slight intensity) + <i>JawDrop</i>)	<i>TongueBackDorsalVelar</i> + <i>LowerLipDepressor</i> + <i>LipTightener</i> + <i>EyesLookUp</i> + <i>LipStretcher</i> + <i>TongueTipAlveolar</i> + <i>LipSuckB</i> + <i>UpperLidRaiser</i> + <i>TongueMidDorsalPalate</i> + <i>ChinRaiserT</i>	<i>LipSuckT</i> + <i>TongueBackDorsalVelar</i> + <i>CheekRaiser</i> + <i>LipSuckB</i> + <i>LipsToward</i> + <i>LipTightener</i> + <i>JawThrust</i> + <i>CheekPuff</i> + <i>UpperLidRaiser</i>
Happiness (<i>CheekRaiser</i> + <i>LipCornerPuller</i>)	<i>LipSuckB</i> + <i>LipsToward</i> + <i>ChinRaiserT</i> + <i>LipSuckT</i> + <i>JawThrust</i> + <i>TongueBackDorsalVelar</i> + <i>CheekRaiser</i>	<i>LowerLipDepressor</i> + <i>TongueTipInterdental</i> + <i>LipFunnelerB</i> + <i>TongueRetreat</i> + <i>TongueBackDorsalVelar</i> + <i>JawThrust</i> + <i>CheekRaiser</i> + <i>CheekPuff</i> + <i>TongueMidDorsalPalate</i> + <i>TongueFrontDorsalPalate</i>
Anger (<i>BrowLowerer</i> + <i>UpperLipRaiser</i> + <i>LidTightener</i> + <i>LipTightener</i>)	<i>CheekRaiser</i> + <i>TongueFrontDorsalPalate</i> + <i>LipCornerDepressor</i> + <i>LipsToward</i> + <i>LowerLipDepressor</i> + <i>LipStretcher</i>	<i>JawThrust</i> + <i>EyesLookRight</i> + <i>LipSuckB</i> + <i>ChinRaiserB</i> + <i>LipFunnelerB</i> + <i>LipCornerDepressor</i> + <i>LipPucker</i> + <i>TongueBackDorsalVelar</i> + <i>LipSuckT</i>
Disgust (<i>NoseWrinkler</i> + <i>LipCornerDepressor</i> + <i>ChinRaiser</i>)	<i>TongueRetreat</i> + <i>LipTightener</i> + <i>CheekRaiser</i> + <i>CheekPuff</i> + <i>UpperLipRaiser</i> + <i>LipPucker</i>	<i>TongueFrontDorsalPalate</i> + <i>LipTightener</i> + <i>TongueMidDorsalPalate</i> + <i>JawSidewaysLeft</i> + <i>MouthLeft</i> + <i>TongueOut</i> + <i>TongueBackDorsalVelar</i>
Sadness (<i>InnerBrowRaiser</i> + <i>BrowLowerer</i> + <i>LipCornerDepressor</i>)	<i>UpperLipRaiser</i> + <i>TongueBackDorsalVelar</i> + <i>TongueFrontDorsalPalate</i> + <i>LipFunnelerB</i> + <i>LowerLipDepressor</i> + <i>CheekRaiser</i> + <i>LipSuckB</i> + <i>TongueRetreat</i>	<i>MouthLeft</i> + <i>LipsToward</i> + <i>TongueOut</i> + <i>CheekRaiser</i> + <i>LipFunnelerB</i> + <i>TongueBackDorsalVelar</i> + <i>LipSuckT</i>
Fear (<i>InnerBrowRaiser</i> + <i>OuterBrowRaiser</i> + <i>BrowLowerer</i> + <i>UpperLidRaiser</i> + <i>LidTightener</i> + <i>LipStretcher</i> + <i>JawDrop</i>)	<i>TongueTipAlveolar</i> + <i>JawDrop</i> + <i>EyesLookRight</i> + <i>JawThrust</i> + <i>CheekRaiser</i> + <i>LipSuckT</i>	<i>JawDrop</i> + <i>CheekSuck</i> + <i>LipPressor</i> + <i>CheekPuff</i>

AU – appears on the Ekman’s list of AUs and either on our natural or acted list of emotions

AU – appears on both our natural and acted emotions list of AUs

AU – appears in all three columns

For Anger, Ekman identified *BrowLowerer*, *UpperLipRaiser*, *LidTightener*, and *LipTightener* as relevant AUs. Strikingly, none of these appear in our model-selected AUs. Instead, our study highlighted *LipCornerDepressor* in both natural and acted expressions of Anger. Natural expressions featured lip movements and consistently engaged *LipCornerDepressor* and *LipStretcher*. Acted expressions also included *LipCornerDepressor* but added more expressive mouth movements, such as *LipPucker*, *LipSuckB*/*LipSuckT*, and *LipFunnelerB*. While neither type of expression fully aligns with Ekman’s, both show partial overlap through shared lip actions.

Regarding Disgust, Ekman proposed *NoseWrinkler*, *LipCornerDepressor*, and *ChinRaiser* as key AUs. While our study did not highlight any of these AUs exactly, both versions pointed to *LipTightener* as a shared feature. The natural model emphasized *CheekRaiser* alongside *LipTightener*, whereas the acted model did not assign significant weight to Ekman’s proposed units. Both expressions diverged from Ekman’s indicators, with *NoseWrinkler* notably absent. Acted Disgust prominently featured tongue movements, possibly reflecting exaggerated rejection signals by our study participants.

In the case of Sadness, Ekman identified *InnerBrowRaiser*, *BrowLowerer*, and *LipCornerDepressor*. Our models instead pointed to *CheekRaiser*, *LipFunnelerB*, and *TongueBackDorsalVelar* as shared AUs for both natural and acted Sadness. Regarding Sadness, the natural model prioritized *UpperLipRaiser*, *LowerLipDepressor*, and

LipSuckB, while the acted model focused on *LipPressor*, *LipFunnelerB*, and *LipSuckT*, revealing a marked difference in lip movements between the two expression types.

Finally, for Fear, Ekman listed several brow and lid movements along with *JawDrop*, which was given a high importance by both our natural and acted models. Our natural model predicted *JawThrust* as an additional jaw feature, while the acted model identified *CheekSuck*. Although the natural model selected *CheekRaiser*, and *LipSuckT*, the acted model highlighted *LipPressor*, and *CheekPuff*. The limited overlap between these features and the poor performance of Fear detection models suggest both a data limitation and a human tendency to mask Fear [18].

Figure 3 provides visual examples of the natural and acted expressions for each of the six basic emotions. For each emotion, the image on the left shows a natural expression elicited through passive video viewing, while the right image shows the same participant performing an acted version of that emotion. These visualizations illustrate the consistent differences between natural and acted expressions, i.e., acted emotions exhibit more intense and anatomically broader facial movements, while natural expressions tend to be subtler and more spatially localized.

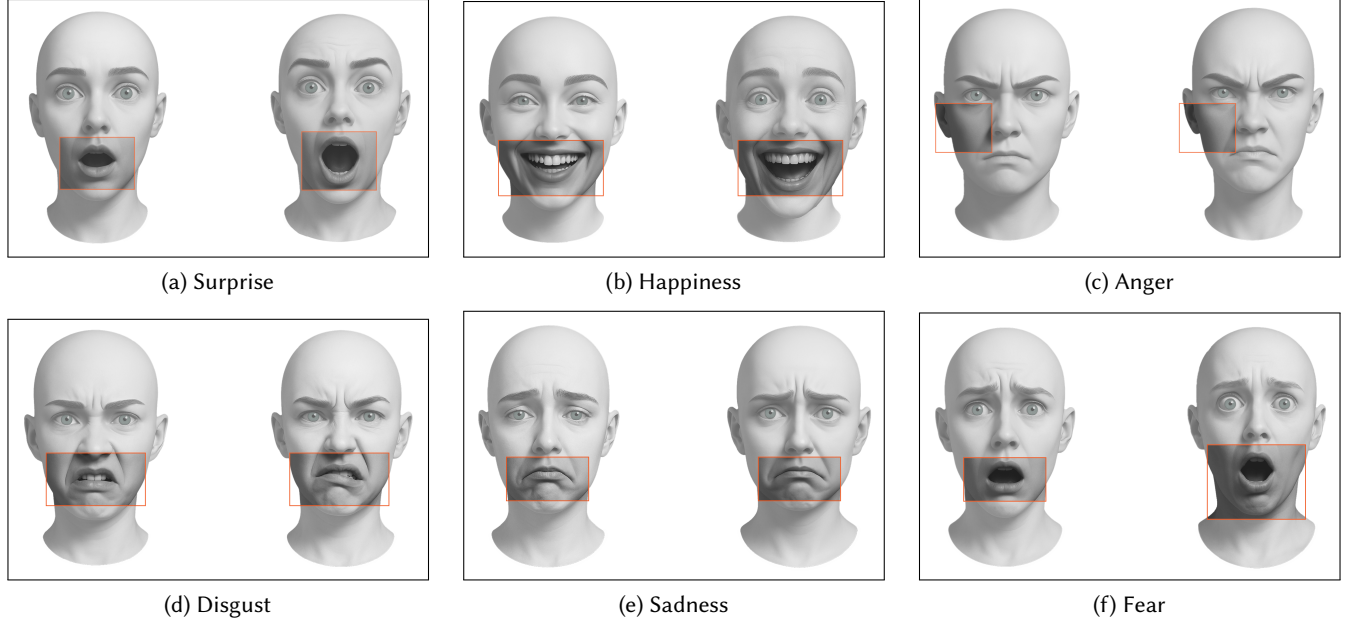


Figure 3: Natural (left) and acted (right) facial expressions for the six basic emotions, with key facial regions highlighted.

5 RQ2: ARE EMOTIONS MORE DIFFICULT TO RECOGNIZE BASED ON NATURAL FACIAL EXPRESSIONS THAN ACTED FACIAL EXPRESSIONS?

Answering our RQ2 is relevant for developing practical models that detect emotions in VR, as it highlights the importance of training on natural emotional expressions and the risk of overestimating model performance when evaluating only on acted data. To realistically address RQ2 in a way that simulates practical use where the model continuously detects emotions over time as the user is wearing the VR headset, we used sequential sequence-to-one models trained to classify emotions from short windows of AU readings. Based on the gathered results and the testing performed on our sequence-to-one models, we then performed statistical hypothesis testing to determine whether our hypothesis holds or not.

5.1 Data Preprocessing

Following the same general procedure used for the AU selection process, we began by filtering out any recordings in which participants did not self-report experiencing the target emotion. These segments were excluded to ensure that each labeled sequence corresponded to a verified emotional response. Next, we labeled sequences of frames in which participants were experiencing the target emotion. Since this analysis used sequence-to-one models, we assigned labels at the level of frame sequences rather than individual frames (i.e., single AU snapshots at a given time). To do this, we first determined whether individual frames potentially contained the target emotion based on their AU values.

For the neutral class, we labeled a frame as potentially neutral if all AUs had values less than or equal to 0.25. For all other emotions,

we labeled a frame as potentially containing the emotion if any AU exceeded a value of 0.25. We chose this thresholding scheme by first examining the distribution of AU values across the entire dataset, observing that 0.25 served as a natural split between background noise (i.e., a neutral facial expression) and meaningful emotions.

We then applied a sliding window approach, examining sequences of five consecutive frames. If at least three out of the five frames in a window were labeled as potentially containing the target emotion, we labeled the entire 5-frame sequence as positive for that emotion. Once a sequence was labeled, we skipped ahead by five frames to avoid overlap between training samples. If the sequence was not labeled, we advanced the window by one frame and repeated the check. This procedure continued until the end of each recording.

After labeling all sequences across the dataset, similarly to what we did when training our logistic regression models, we used a stratified 5-fold cross-validation averaged across 10 runs to evaluate the models.

5.2 Model

We experimented with different Recurrent Neural Network (RNN) and Convolutional Neural Network (CNN) based models to predict the emotion a person might be feeling based on a sequence of AU readings. More specifically, we explored combinations of Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) layers from RNNs, and 1D Convolutional (Conv1D) and Temporal Convolutional Network (TCN) layers from CNNs. For classification, we added a dense layer with 7 units to all models and used a *softmax* activation function to output the model's predicted probability for each emotion. For each prediction, we selected the emotion with the highest probability as the model's output.

Table 2: Emotion classification performance (F1-score) across training and testing conditions, e.g., Natural → Natural denotes training on Natural emotion data and testing on Natural emotion data.

Emotion	Natural → Natural	Acted → Acted	Natural → Acted	Acted → Natural
Anger	0.52	0.72	0.05	0.15
Disgust	0.85	0.78	0.36	0.27
Fear	0.52	0.62	0.13	0.20
Happiness	0.72	0.85	0.18	0.10
Neutral	0.91	0.95	0.91	0.80
Sadness	0.67	0.85	0.35	0.12
Surprise	0.46	0.79	0.07	0.08
Overall	0.70*	0.80*	0.28	0.22

* Significant difference between Natural→Natural and Acted→Acted (Wilcoxon signed-rank test, $p < 0.05$).

All time series models were trained using Stochastic Gradient Descent as the optimizer. We used sparse categorical cross-entropy, weighted by class frequency, as the loss function across all models. We adopted a stratified 5-fold cross-validation procedure with 80% of the data for training and 20% for testing. Models were trained for 30 epochs with a batch size of 20, using data from all emotion classes. To mitigate overfitting, we applied a dropout rate of 0.3. The convolutional layers used 64 filters with a kernel size of 5, and padding was applied to preserve input shape across convolutions.

Our preliminary experiments showed that the model consisting of a single TCN layer outperformed all other architectures. We then performed hyperparameter tuning on this model to optimize its configuration. A number of hyperparameters were specific to the TCN layer, such as the number of filters, dropout rate, and kernel size. Others controlled the inclusion of additional dense layers following the TCN and the number of units in those layers. The remaining parameters applied more generally to the model, including batch size, number of training epochs, activation function for intermediate layers, and the learning rate and momentum used by the optimizer. For the natural emotion dataset, the optimal TCN configuration consisted of 128 filters, a kernel size of 4, no additional dense layers, a dropout rate of 0.2, a batch size of 20, 30 training epochs, a ReLU activation function, a learning rate of 0.01, and momentum set to 0.95. For the acted emotion dataset, the best configuration used the same number of filters and training parameters but included one dense layer with 256 units, a kernel size of 5, and a higher dropout rate of 0.3.

5.3 Metrics

We used F1-score since it is as comprehensive metric that combines both precision and recall into a single value by computing their harmonic mean. Precision measures the proportion of true positive predictions among all predicted positives, given by $\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}}$. Recall, on the other hand, captures the proportion of true positives identified out of all actual positives, defined as $\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}}$. The F1-score balances these two by calculating $2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$, providing a more informative metric when dealing with imbalanced classes. In multi-class or imbalanced classification tasks, a class-weighted F1-score can be used, where each class's F1-score is weighted by its support (i.e., the number of true instances in that

class), allowing the overall metric to reflect the importance of each class in the dataset. To rigorously evaluate difference in the F1 scores of the natural model and the acted model, we then conducted statistical significance testing using Wilcoxon's signed-rank test on the models' F1-scores. We chose this non-parametric test as it does not assume normality of the underlying score distributions, making it well-suited for evaluating paired model outputs.

5.4 Results

We report the performance of the models for different train and test settings of natural and acted data in Table 2. Based on these results, we make the following observations:

In our within-domain (i.e., Natural → Natural and Acted → Acted) evaluations, models trained and tested on acted data consistently outperformed those trained and tested on natural expressions. The overall F1-score for acted data was 0.80, while natural data reached only 0.70, indicating that standard FER models overestimate performance when evaluated solely on acted inputs. For all emotion classes except *Disgust*, acted expressions yielded higher F1-scores, e.g., 0.85 for Happiness and 0.79 for Surprise, compared to 0.72 and 0.46 in the natural setting. For *Disgust*, the natural model yielded a F1-score of 0.85, compared to 0.78 for the acted model. The acted expressions usually had extreme facial dynamics, as discussed in RQ1, which made them easier for the models to learn. Natural expressions, in contrast, were more heterogeneous and subtle, strongly reducing classification performance.

Cross-domain (i.e., Natural → Acted and Acted → Natural) performance was considerably worse than within-domain when testing both cases. For instance, the F1-score for Surprise, dropped to 0.08 when the model was trained on acted expressions and tested on natural, and to 0.7 when the domains were reversed. Similar degradation was observed across most emotion classes. This lack of transferability highlights a significant domain shift between natural and acted emotional expressions. Models trained on one do not learn patterns that generalize well to the other. These results caution against relying on acted datasets for training systems meant to operate in naturalistic VR environments.

These results indicate systematic differences between natural and acted emotions (Table 1), with model performance suggesting that natural emotions are inherently more challenging to predict. Finally, the Wilcoxon's signed-rank test yielded a p-value of 0.0469, which falls below the conventional threshold of $p < 0.05$. We

therefore conclude that natural emotions are significantly more difficult to recognize than acted emotions.

6 IMPLICATIONS

The results of this study affirmatively answered the two research questions. For RQ1, we showed that natural and acted emotions in VR differ in systematic ways. Natural expressions are subtler, more heterogeneous, and spatially localized, while acted expressions are exaggerated and involve broader muscle activity. These findings establish a clear domain gap between natural and acted emotions, which indicates that acted datasets cannot serve as reliable substitutes for natural expressions. For RQ2, we demonstrated that natural emotions are significantly more difficult to recognize than acted ones. Models trained and evaluated on acted data achieved higher F1-scores, but they generalized poorly when applied to natural expressions. This confirms that standard FER models overestimate performance when validated only on acted data and that robust recognition in VR requires models trained explicitly on natural expressions.

The implications of these findings are relevant to several groups. For researchers in affective computing, the results highlight the need to prioritize naturalistic data in both dataset construction and model evaluation. Future work must address the complexity of natural expressions by leveraging temporal modeling and multimodal signals, rather than relying predominantly on acted datasets. For VR system designers and developers, the findings show that reliable affect-adaptive applications cannot be built on acted-only data. Pipelines must incorporate naturalistic training material to support dependable real-time inference in interactive VR settings. Finally, for application domains such as gaming, mental health, or training simulations, relying on acted datasets risks misinterpretation of subtle user states, which can erode trust or reduce effectiveness.

7 THREATS TO VALIDITY

Several limitations may impact the interpretation of our findings. We categorize and list each of them below.

Construct validity concerns how well the study's observations align with the theoretical constructs of emotion. We use self-reported labels to define the ground truth for natural emotions. These reports are subjective and depend on participants' awareness and interpretation of their own emotional states. The mapping from Action Unit (AU) data to discrete emotion labels assumes consistent physiological expression across participants, which is unlikely. We mitigate this by discarding AU data for any emotion not explicitly reported by the participant. This ensures that each sequence used in training and evaluation corresponds to an acknowledged emotional state. We note that the captured AU data is continuous, but the self-reports after each recording were collected using a Likert scale. The temporal mismatch is a limitation, but it follows prior works that use post-recording ratings to validate continuous emotion data [3, 32].

Internal validity concerns whether the study design introduces unintended bias. The natural condition always precedes the acted condition, which introduces order effects. Participants may unintentionally reproduce features of their natural expressions when asked to act, which would reduce the observed difference between

the two conditions. We rely entirely on the Meta Quest Pro's AU tracking API. The accuracy of its facial capture pipeline is proprietary and not externally validated. Factors like headset fit or face shape could systematically affect AU values across participants.

External validity concerns how well the results generalize. Most participants are university students, limiting demographic and cultural diversity. The emotion elicitation uses passive video watching in a lab, which does not reflect how emotions emerge in real-world or interactive VR experiences. The poor performance of models across domains (e.g., training on acted, testing on natural) shows that the learned representations do not transfer. These limitations reduce the applicability of the results to deployment scenarios unless training data matches the target use context. Additionally, we did not record participants' prior VR experience. Novelty effects, especially among first-time VR users, may influence both emotional engagement and expressiveness.

8 CONCLUSIONS AND FUTURE WORK

This paper presents a method for recognizing facial emotions in VR using AU data collected from the Meta Quest Pro headset. We compare natural and acted emotional expressions from the same participants and show that they activate different AUs with distinct temporal patterns. Models trained on acted data consistently perform better than those trained on natural expressions. Cross-domain generalization is poor, confirming that natural and acted expressions are not interchangeable for training.

As next steps, we will release a version of the recognition model trained jointly on both natural and acted emotion data. This version will improve robustness across expression types. We also plan to expand the dataset with more diverse participants and evaluate the model in task-based VR settings to support affect-adaptive applications. In addition, we plan to extend our study of natural emotions to different modalities like speech and gestures, to help with better classifying subtler emotions such as Fear. Future studies can also consider participants' non-identifiable demographic data (e.g., age ranges, years of VR experience) into account, as it may affect how they express emotions in virtual environments. To achieve better alignment between continuous face readings and self-reported validation responses, future works can explore continuous self-report methods, such as timestamped tagging.

ACKNOWLEDGMENTS

We would like to thank Won Bin Lee, Ha Rin Yoo, Ha Jung Woo, Ji Seo Myung, Song Lee, Yu Jin Choi, Seung, Pyo Kang, and Se Jin Jang for their valuable feedback and support during the development of this work.

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